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Milnor *Morse Theory* Chapter 14: Jacobi Fields: The Null Space of E_{**} .

Jacobi fields

Recall the second variation formula

$$\frac{1}{2} \frac{\partial^2 E}{\partial u_1 \partial u_2}(0, 0) = - \sum_t \left\langle W_2(t), \Delta_t \frac{\partial W_1}{\partial t} \right\rangle - \int_0^1 \left\langle W_2, \frac{\partial^2 W_1}{\partial t^2} + \mathcal{R} \left(W_1, \frac{\partial \omega}{\partial t} \right) \frac{\partial \omega}{\partial t} \right\rangle dt.$$

We know that a geodesic is a critical point, but this alone does not imply it is minimal.

This motivates us to define: A vector field J along ω is called a **Jacobi field** if it satisfy the Jacobi differential equation

$$\frac{\partial^2 J}{\partial t^2} + \mathcal{R} \left(J, \frac{\partial \omega}{\partial t} \right) \frac{\partial \omega}{\partial t} = 0. \quad (\star)$$

This is a second order linear ODE, so the solution is uniquely determined by the initial conditions $J(0), \frac{\partial J}{\partial t}(0) \in T_{\omega(0)}M$, and all solutions are smooth.

Let $p = \omega(0), q = \omega(1)$ be two points on the geodesic γ , $T\Omega_\omega$ denotes the vector fields along ω vanishing at p, q .

Recall the null space of $E_{**} : T\Omega_\omega \times T\Omega_\omega \rightarrow \mathbb{R}$ is the kernel of $\tilde{E} : T\Omega_\omega \rightarrow T\Omega_\omega^*, W_1 \mapsto E_{**}(W_1, \cdot)$. The nullity ν of E_{**} is the dimension of the null space and E_{**} is degenerate if $\nu > 0$.

Theorem: A vector field $W_1 \in T\Omega_\omega$ belongs to the null space iff W_1 is a Jacobi field.

Proof: If J is a Jacobi field vanishing at p, q , then $J \in T\Omega_\omega$ and the second variation formula implies

$$-\frac{1}{2} E_{**}(J, W_2) = \sum \langle W_2(t), 0 \rangle + \int_0^1 \langle W_2, 0 \rangle dt = 0.$$

Hence J belongs to the null space.

Conversely, suppose W_1 is in the null space. Choose a subdivision $0 = t_0 < \dots < t_k = 1$ such that $W_1|_{[t_{i-1}, t_i]}$ is smooth, and let $f : [0, 1] \rightarrow [0, 1]$ be smooth such that $f(t_i) = 0$ and otherwise $f(x) > 0$. Let

$$W_2(t) = f(t) \left(\frac{\partial^2 W_1}{\partial t^2} + \mathcal{R}(W_1, V)V \right).$$

Then

$$0 = -\frac{1}{2} E_{**}(W_1, W_2) = \int_0^1 f(t) \left\| \frac{\partial^2 W_1}{\partial t^2} + \mathcal{R}(W_1, V)V \right\|^2 dt.$$

Therefore W_1 satisfy $(\star)$ on $x \neq t_i$.

Now let $W_2' \in T\Omega_\omega$ be a field such that $W_2'(t_i) = \Delta_{t_i} \frac{\partial W_1}{\partial t}$ for $i = 1, 2, \dots, k-1$, then

$$0 = -\frac{1}{2}E_{**}(W_1, W_2') = \sum_{i=1}^{k-1} \|\Delta_{t_i} \frac{\partial W_1}{\partial t}\|^2$$

so $\frac{\partial W_1}{\partial t}$ is continuous. By ODE theory, a C^1 solution of $(\star)$ is smooth. $\square$

It follows that the nullity ν is finite, since there are at most n linearly independent Jacobi fields along ω given the boundary condition $J(0) = 0$. Actually $\nu < n$, since we can find $J(0) = 0$ but $J(1) \neq 0$: Let $J(t) = tV(t)$ where $V = \frac{\partial \omega}{\partial t}$ is the velocity, then $\frac{\partial J}{\partial t} = V + t \frac{\partial V}{\partial t} = V$ since $\frac{\partial V}{\partial t} = 0$. Hence $\frac{\partial^2 J}{\partial t^2} = 0$. Also, $\mathcal{R}(J, V)V = t\mathcal{R}(V, V)V = 0$. Therefore J is a Jacobi field with $J(0) = 0$ but $J(1) = V(1) \neq 0$.

Example: Suppose M is "flat" such that $\mathcal{R} \equiv 0$, then the Jacobi equation becomes $\frac{\partial^2 J}{\partial t^2} = 0$. Suppose $J = \sum f^i P_i$ where P_i are parallel along ω , then the equation is $\frac{\partial^2 f_i}{\partial t^2} = 0$ so each f_i is linear. Hence any Jacobi field has at most one zero, so E_{**} is non-degenerate.

Let α be a variation of ω , (not necessarily keeping endpoints fixed), such that $\alpha(u, \cdot)$ is a geodesic for all $u \in (-\varepsilon, \varepsilon)$ and $\alpha(0, \cdot) = \omega$.

Lemma: If α is such a variation of ω through geodesics, then the variation vector field $W(t) = \frac{\partial \alpha}{\partial u}(0, t)$ is a Jacobi field along ω .

Proof: Note that

$$\begin{aligned} 0 &= \frac{\partial}{\partial u} \left(\frac{\partial}{\partial t} \frac{\partial \alpha}{\partial t} \right) = \frac{\partial}{\partial t} \frac{\partial}{\partial u} \frac{\partial \alpha}{\partial t} + \mathcal{R} \left(\frac{\partial \alpha}{\partial u}, \frac{\partial \alpha}{\partial t} \right) \frac{\partial \alpha}{\partial t} \\ &= \frac{\partial^2}{\partial t^2} \frac{\partial \alpha}{\partial u} + \mathcal{R} \left(\frac{\partial \alpha}{\partial u}, \frac{\partial \alpha}{\partial t} \right) \frac{\partial \alpha}{\partial t}. \end{aligned}$$

Hence evaluate at $(u, t) = (0, t)$ we infer W is a Jacobi field. $\square$

Thus one way of obtaining Jacobi fields is moving geodesics around (an exponential map $\alpha(u, t) = \exp_p tv(u)$). The converse is also true:

Lemma: Every Jacobi field along a geodesic $\omega : I \rightarrow M$ may be obtained by a variation of ω through geodesics.

Proof: If J is a Jacobi field along ω with $J(0) = 0$, then let $w = \frac{\partial J}{\partial t}(0)$ and $v = \frac{\partial \omega}{\partial t}(0)$. Take $v(s) : I \rightarrow T_{\omega(0)}M$ such that $v(0) = v$, $v'(0) = w$ and let $\alpha(u, t) = \exp_p(tv(u))$, then $W = \frac{\partial \alpha}{\partial u}(0, t)$ is a Jacobi field such that $W(0) = J(0) = 0$ and $\frac{\partial W}{\partial t}(0) = \frac{\partial J}{\partial t}(0) = w$. Hence by uniqueness $W = J$.

Conjugate points

Definition: p, q are **conjugate** along ω if there exists a non-zero Jacobi field J along ω which vanishes for $t = a, b$. The **multiplicity** of p, q is equal to the dimension of all such Jacobi fields as an $\mathbb{R}$ -vector space.

Then the Theorem states that: E_{**} is degenerate iff the end points p, q are conjugate along ω ; and ν is equal to the multiplicity of p, q .

Example: Suppose p, q are antipodal points on $\mathbb{S}^n$, and ω a great circle arc from p to q . Using Lemma, we can rotate the sphere in $n - 1$ directions, fixing p, q , to obtain $n - 1$ linearly independent Jacobi fields. Hence p, q are conjugate of multiplicity $n - 1$, so the nullity ν takes its maximal value.

So conjugation shows how the geodesic fails to be of minimal energy.

More of this topic is discussed in the next chapter.